

Program 1(Single and Compound indexing)

Create collection name student and apply query to find student who have age greater than 12 using indexing.

_id	Name	age
1	ABC	10
2	XYZ	12
3	ABC	15
4	PQR	13
5	XYZ	15
6	ABC	8

Program 2(Partial indexing)

Consider student collection with 6 documents

- 1. Create index on age where age is greater than 15**
- 2. Display student/s whose age is 16.**
- 3. Display student/s whose age is 13.**
- 4. Display student/s whose age is 17.**

Program 3

Consider a collection student having documents like this:

```
[{_id:123433,name: "DDD",age:32},  
{_id:123434,name: "BBB",age:20},  
{_id:123435,name: "AAA",age:10},]
```

Do as directed:

- (1) Create an index & fire a command to retrieve a document having age>15 and name is "BBB". Stats must return values nReturned=1, docExamined=1, stage="IXSCAN". Perform required indexing.
- (2) Create an index on subset of a collection having age>30. Also write a command to get a stats "IXSCAN" for age>30.

Program 4

Create student collection and perform the task as ask below.

```
[
  { _id:1, name:"ABC", city:"Ahmedabad", age:18, marks:75 },
  { _id:2, name:"XYZ", city:"Surat", age:20, marks:82 },
  { _id:3, name:"PQR", city:"Ahmedabad", age:19, marks:68 },
  { _id:4, name:"ABC", city:"Rajkot", age:21, marks:90 },
  { _id:5, name:"XYZ", city:"Ahmedabad", age:20, marks:85 },
  { _id:6, name:"LMN", city:"Surat", age:18, marks:60 }
]
```

Question 1: Simple Index

Create an index on the age field and retrieve all students whose age is 20. Verify the execution statistics.

Question 2: Compound Index

Drop the existing index. Create a compound index on age and name. Retrieve the student whose age is 20 and name is "XYZ". Verify the execution statistics.

Question 3:

Using the same compound index, retrieve all students whose name is "ABC". Verify the execution statistics.

Question 4: Partial Index

Drop the existing compound index. Create a partial index on the marks field for students having marks greater than 80. Retrieve all students whose marks are greater than 80 and verify the execution statistics.

Question 5:

Using the same partial index, retrieve all students whose marks are greater than 70. Verify the execution statistics.